

Retroreflective Optical ISAC Using OFDM and Corner Cube Reflector

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Abstract—A retroreflective optical integrated sensing and communication (RO-ISAC) system using orthogonal frequency division multiplexing (OFDM) and corner cube reflector (CCR) is reported. To accurately model the reflected sensing channel of the RO-ISAC system, both a point source model and an area source model are proposed. Theoretical and experimental results verify the accuracy of the proposed channel models, and the communication and sensing performance of the considered RO-ISAC system is also evaluated.